Review Form of MLHMI 2026

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Paper ID: BH5094

Paper Title: Enhancing Long-Horizon Robot Manipulation with Hierarchical Task Decomposition and Subtask Finetuning

Criteria	Score: 1-5 (1 = Poor, 3 = Average, 5 = Excellent)
Relevance and Significance – 4/5 The topic is clearly aligned with MLHMI: improving robustness in **long-horizon robot manipulation** using **hierarchical task decomposition** and **subtask finetuning** in VLA models. The paper also targets a meaningful practical issue (execution robustness and cascading errors) via task-structure conditioning.	
Originality and Innovation – 3/5 Hierarchical decomposition is not new, but the paper's framing—**oracle subtask prompts conditioning a fixed VLA policy**, plus **subtask-specific finetuning**—is a reasonable and relevant twist for modern VLA settings. The attention-based prompt sensitivity discussion adds a potentially interesting diagnostic angle.	
Methodology – 4/5 Appropriate and partly rigorous:	

Decomposition is controlled via an **oracle** (manual rules), which is reasonable for isolating the effect of explicit task structure. Finetuning setup is described at a high level and the intent is clear. “Split-rubric framework” isn’t cleanly defined as a method artifact vs an evaluation condition (terminology ambiguity).	
Results and Analysis – 3/5 Results are presented with clear success-rate tables for Task 1 and Task 2. Includes attention analysis comparing π0 vs π0.5 prompt sensitivity. The text claims for Task 1 that π0-split improves “from **45.4% to 53.8%**”, but Table I shows π0-split is **35.6% (Baseline)** → **53.8% (Split)**, while **45.4%** belongs to π0-base under Baseline. This reads like an apples-to-oranges or mistaken comparison and undermines the credibility of the headline result unless corrected.	
Discussion and Conclusion – 4/5 The conclusion is consistent with the study design and highlights that benefits depend on language sensitivity. Limitations are explicitly acknowledged; oracle subtasks, limited tasks, no recovery.	
Structure and Organization – 3/5	

The high-level structure is present (method → results → analysis → limitations → conclusion), however duplicated table content and under-defined terms ("rubric") disrupt flow and can confuse readers.	
Clarity of Expression – 4/5 There is unclear definition of "split-rubric framework", visible word-break artifacts (e.g., "analyz- ing", "execu- tion") and a grammar error in acknowledgments	

Recommendation:

Please select one of the following:

☐ Accept as is

✓ Accept with minor revisions

☐ Revise and resubmit

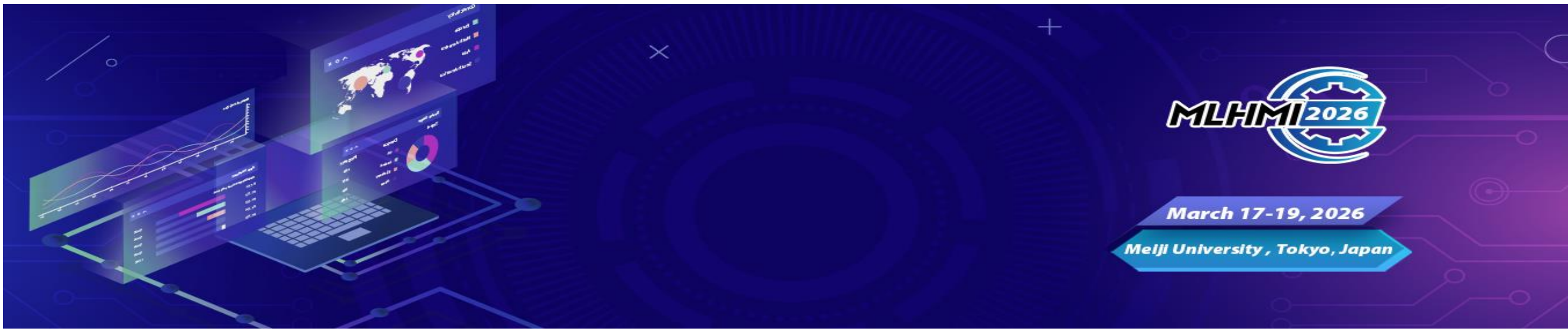
☐ Reject

Additional Comments (Required):

Please provide any feedback, suggestions for improvement, or concerns about the paper.

1. **C**orrect the key comparison and rewrite the main claim so it is strictly within-model and within-rubric (e.g., π0-split baseline → π0-split split should be 35.6% → 53.8%**,** not 45.4% → 53.8%**).**

2. Define "rubric" precisely (is it a prompt template, a decomposition scheme, or an



evaluation condition?) and keep terminology consistent.

3. Add reproducibility details, exact prompts, oracle subtask list + termination, time limits, seeds, finetuning hyperparameters/data splits.